

Sri Sivasubramaniya Nadar College of Engineering, Chennai
(An Autonomous Institution affiliated to Anna University)

Degree & Branch	B.E. Computer Science & Engineering	Semester	VI
Subject Code & Name	UCS2612 – Machine Learning Algorithms Laboratory		
Academic Year	2025–2026 (Even)	Batch	2023–2027
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Experiment 1: Working with Python packages - Numpy, Scipy, Scikit-Learn, Matplotlib

1. Aim and Objective

To comprehensively explore the capabilities of fundamental Python libraries including Numpy, Pandas, Scipy, Scikit-learn, and Matplotlib. The experiment focuses on understanding core concepts in data manipulation, preprocessing techniques, mathematical computing operations, end-to-end machine learning workflows, and effective data visualization strategies. Additionally, the objective includes identifying appropriate machine learning task types for various datasets and determining optimal algorithms based on dataset characteristics and problem requirements.

2. Dataset Description

2.1 Iris Dataset

The Iris dataset is a classic multivariate dataset containing 150 samples of iris flowers, with 50 samples each from three distinct species: *Iris setosa*, *Iris versicolor*, and *Iris virginica*. Each sample is characterized by four continuous features measured in centimeters: sepal length, sepal width, petal length, and petal width. The target variable represents the species classification, making this an ideal dataset for demonstrating multi-class classification algorithms and supervised learning techniques.

2.2 Loan Amount Prediction

This comprehensive financial dataset is designed for regression analysis to predict the sanctioned loan amount based on applicant characteristics. The dataset comprises approximately 30,000 records with diverse features including Customer ID, Name, Age, Annual Income (USD), Requested Loan Amount, Credit Score, and Property Price. The primary objective is to develop a predictive model that accurately estimates the loan amount likely to be approved by financial institutions based on the applicant's financial profile and demographic information.

2.3 Predicting Diabetes

The Diabetes dataset, commonly known as the PIMA Indians Diabetes Database, is utilized for binary classification to predict diabetes occurrence based on diagnostic measurements. This dataset contains 768 samples with eight critical medical features: number of Pregnancies, Glucose concentration, Blood Pressure (mm Hg), Skin Thickness (mm), Insulin level (μ U/ml), Body Mass Index (BMI), Diabetes Pedigree Function (genetic influence), and Age (years). The binary target variable 'Outcome' indicates the presence (1) or absence (0) of diabetes, making it suitable for evaluating classification algorithms in medical diagnosis scenarios.

2.4 Classification of Email Spam

The Spambase dataset is a widely-used benchmark for binary text classification, specifically designed to distinguish between legitimate emails (ham) and unsolicited spam messages. The dataset consists of 4,601 instances characterized by 57 continuous features representing normalized word and character frequencies. These features include frequency counts of spam-indicative words such as 'free' and 'money', as well as special character frequencies like '!' and '\$'. The binary target variable 'spam' (1 for spam, 0 for ham) enables the evaluation of text classification and natural language processing techniques.

2.5 Handwritten Character Recognition / MNIST

The MNIST dataset (Modified National Institute of Standards and Technology database) represents a cornerstone benchmark in computer vision and pattern recognition. It comprises 60,000 training images and 10,000 testing images of handwritten digits (0-9). Each image is represented as a 28×28 pixel grayscale matrix, resulting in 784 features per sample with pixel intensity values ranging from 0 to 255. This multi-class classification problem serves as a standard evaluation metric for image processing algorithms, neural networks, and deep learning architectures.

3. Exploratory Data Analysis and Visualization

3.1 Iris Dataset

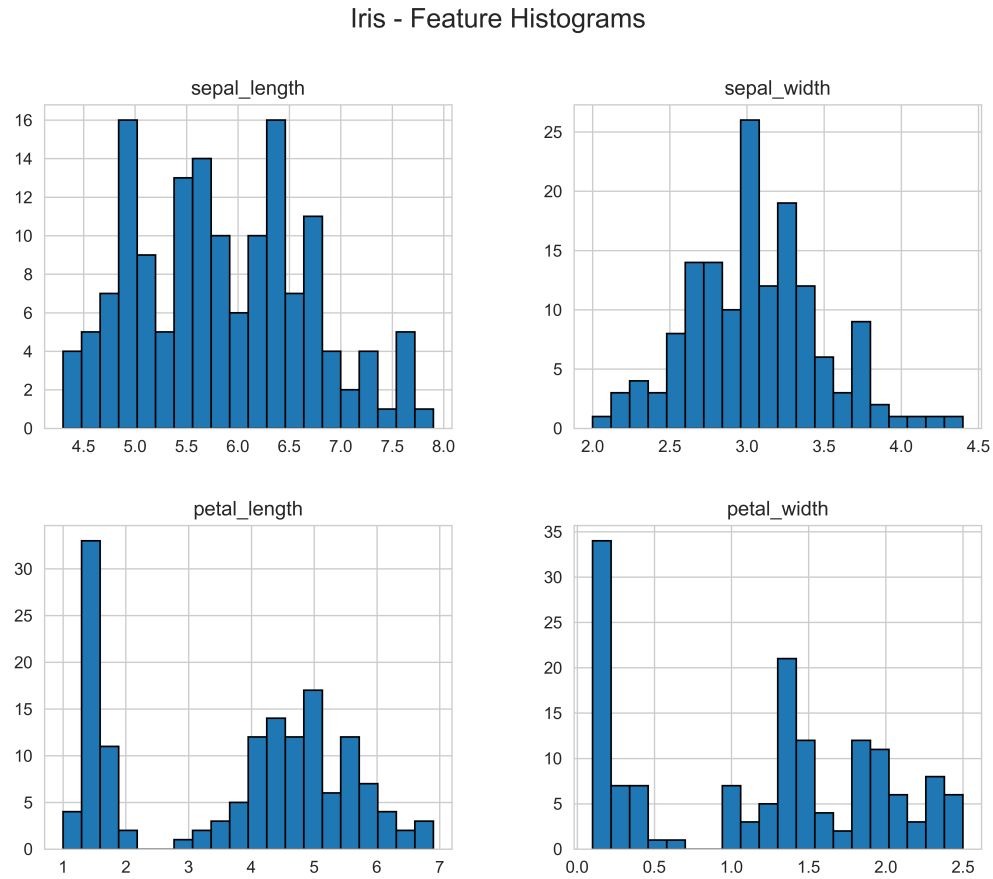


Figure 1: Feature Histograms - Iris Dataset

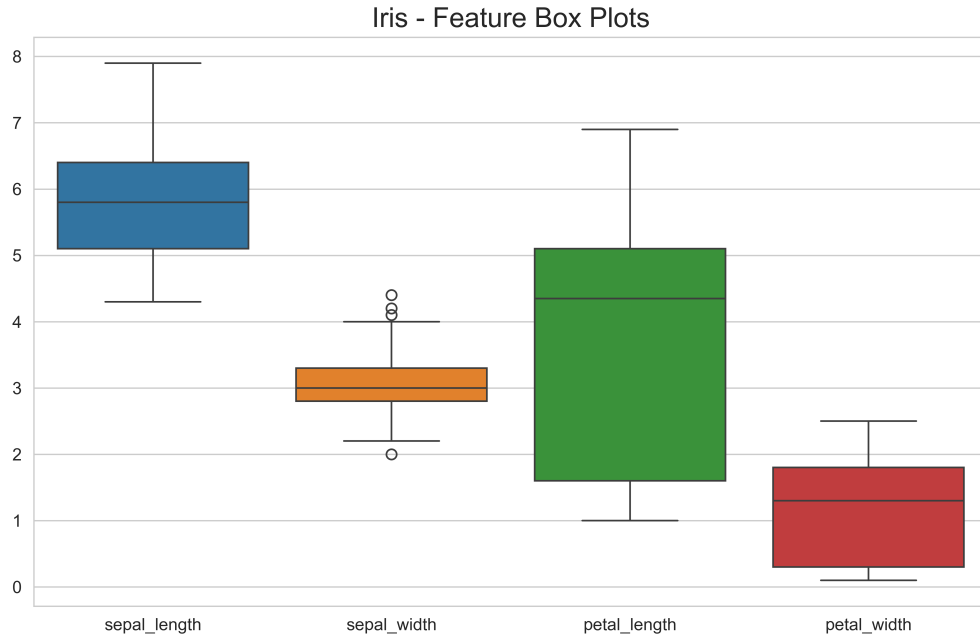


Figure 2: Feature Box Plots - Iris Dataset

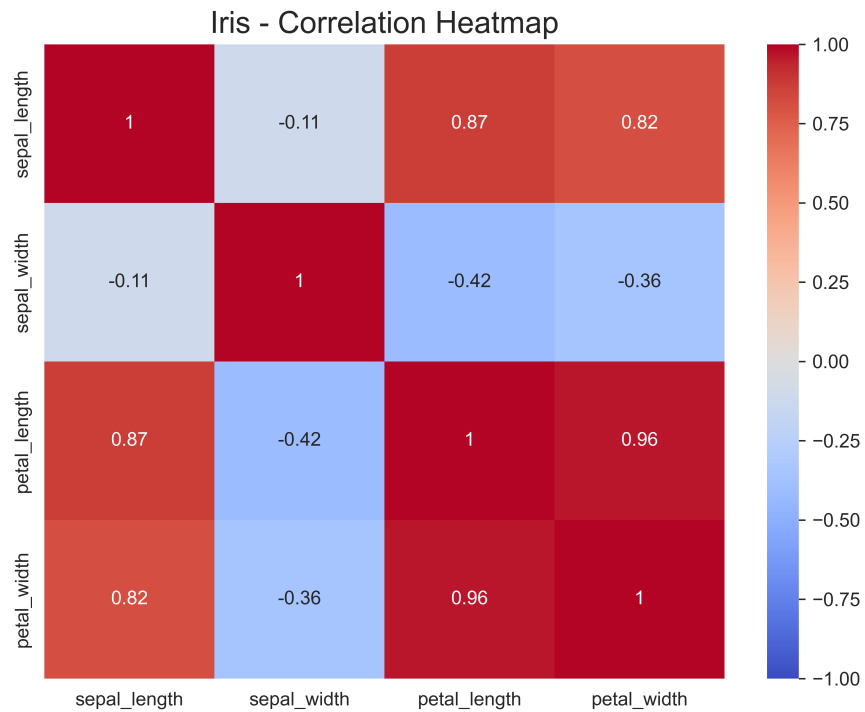


Figure 3: Correlation Heatmap - Iris Dataset

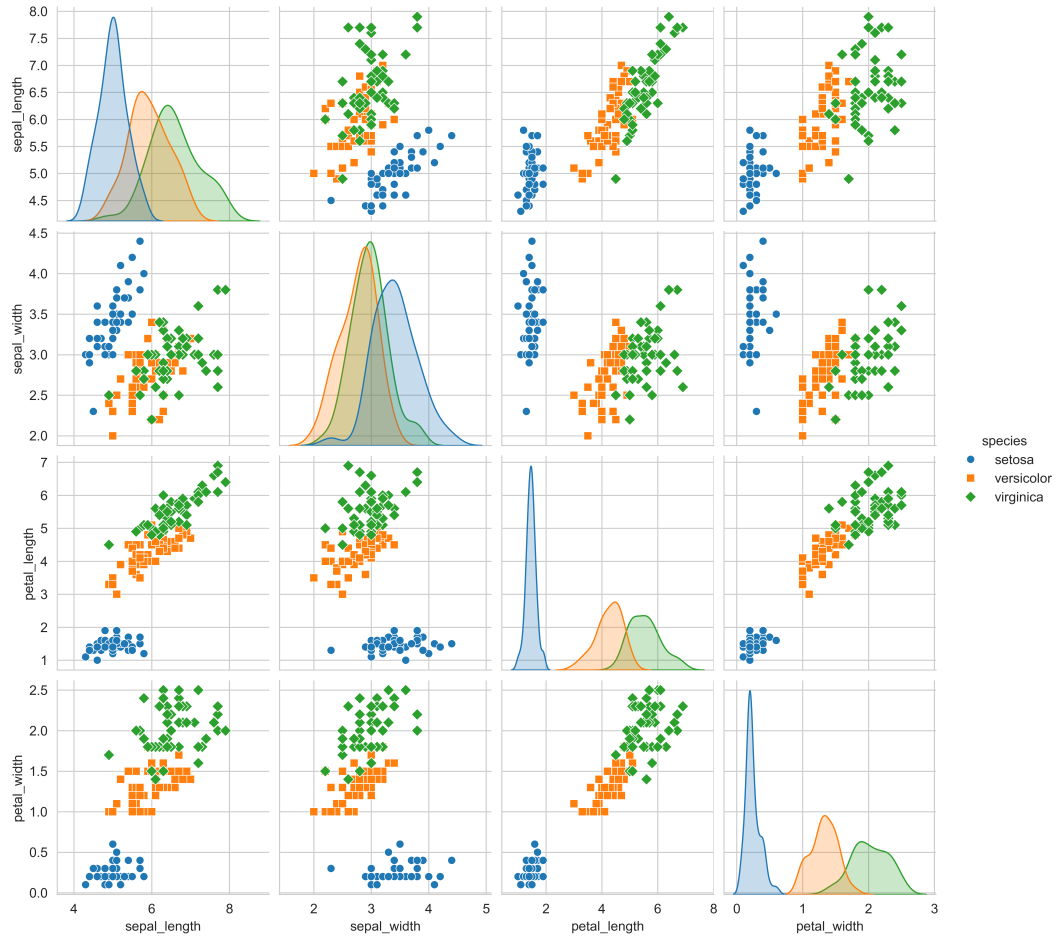


Figure 4: Pair Plot Classification - Iris Dataset

3.2 Loan Amount Prediction

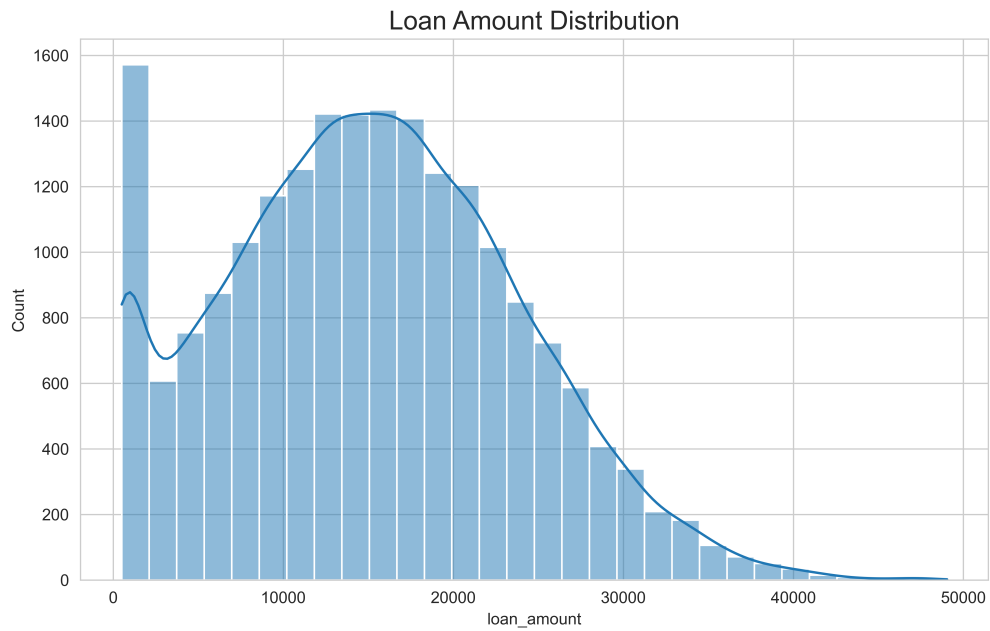


Figure 5: Target Variable Distribution - Loan Amount

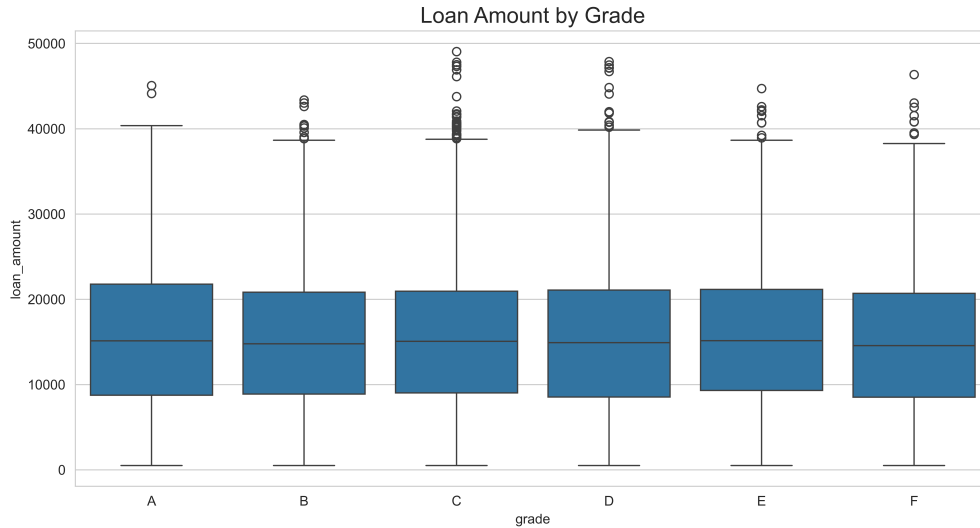


Figure 6: Box Plot by Grade - Loan Amount

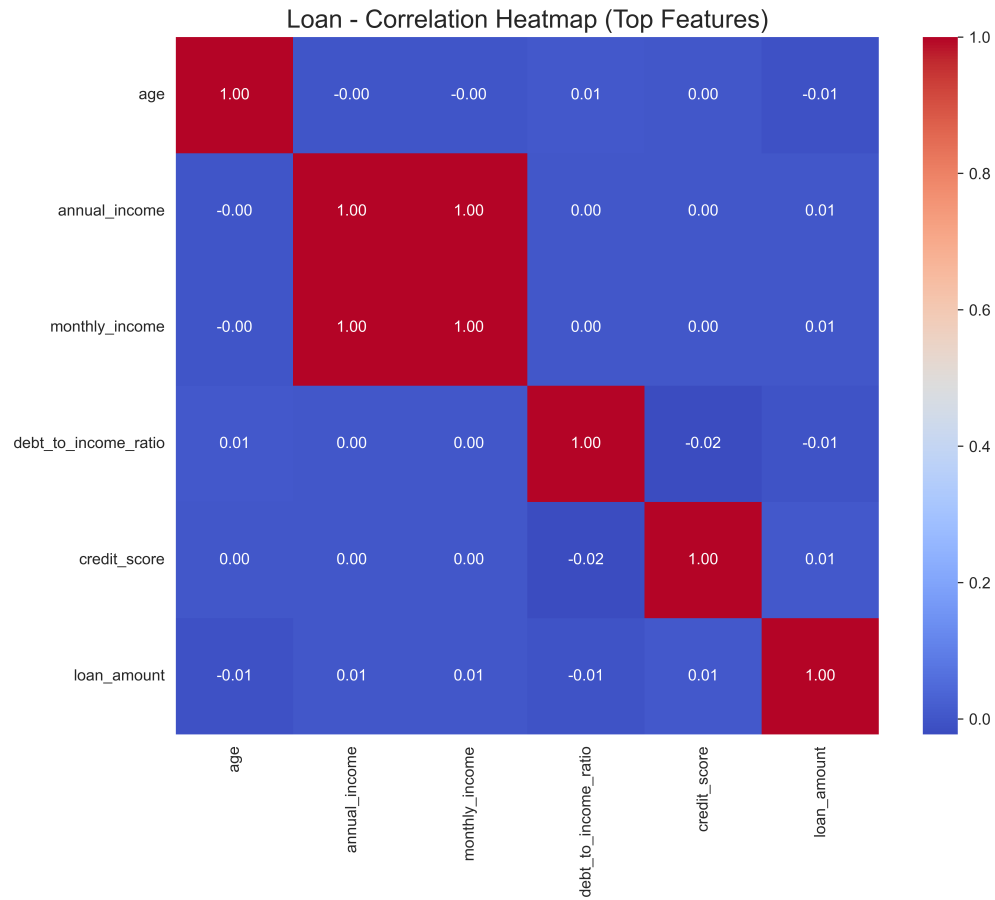


Figure 7: Feature Correlation Heatmap - Loan Amount

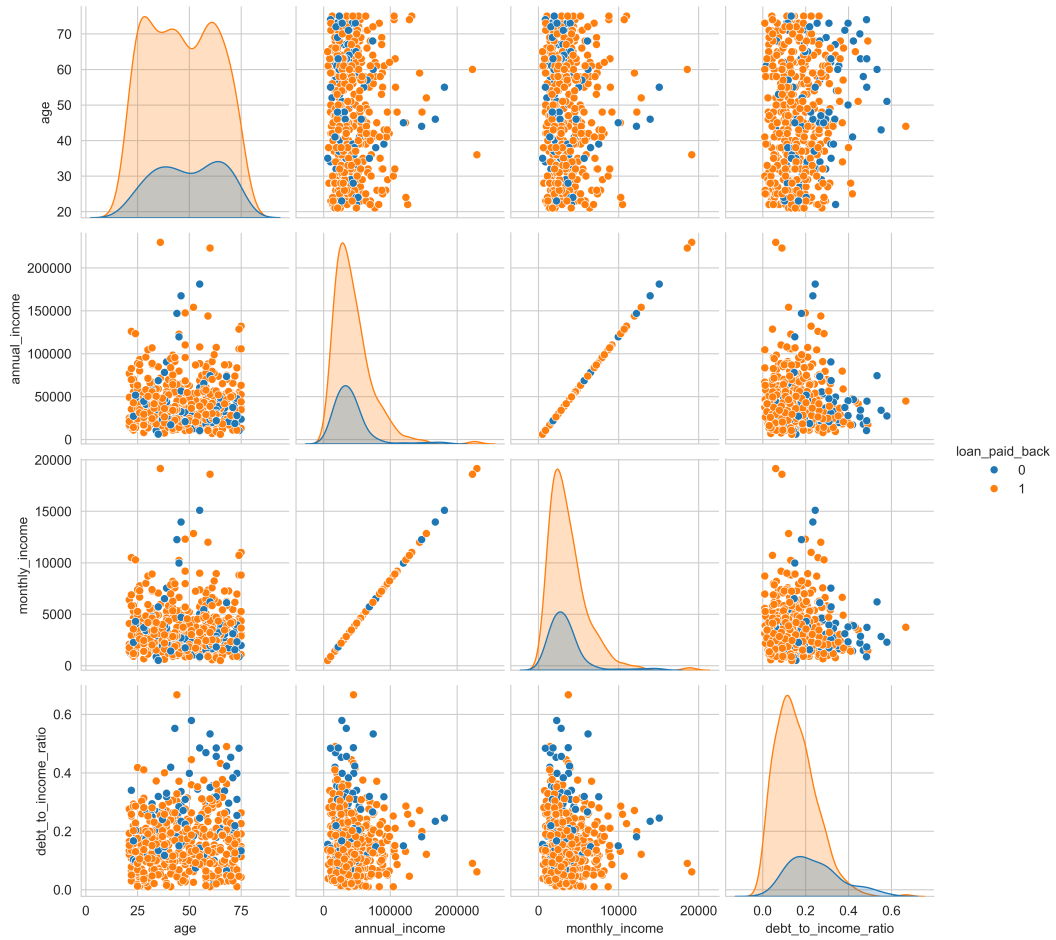


Figure 8: Pair Plot (Subset) - Loan Amount

3.3 Predicting Diabetes

Diabetes - Feature Histograms

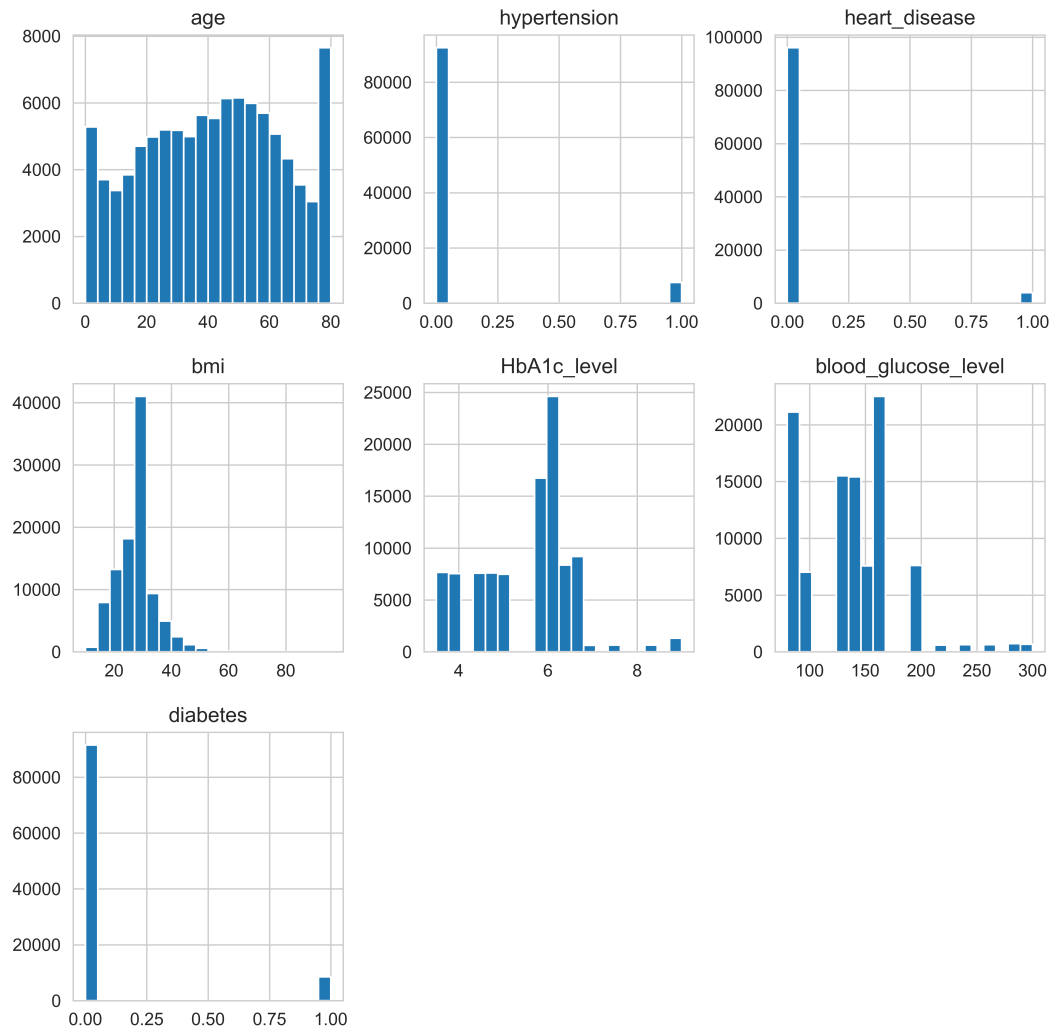


Figure 9: Feature Histograms - Diabetes Dataset

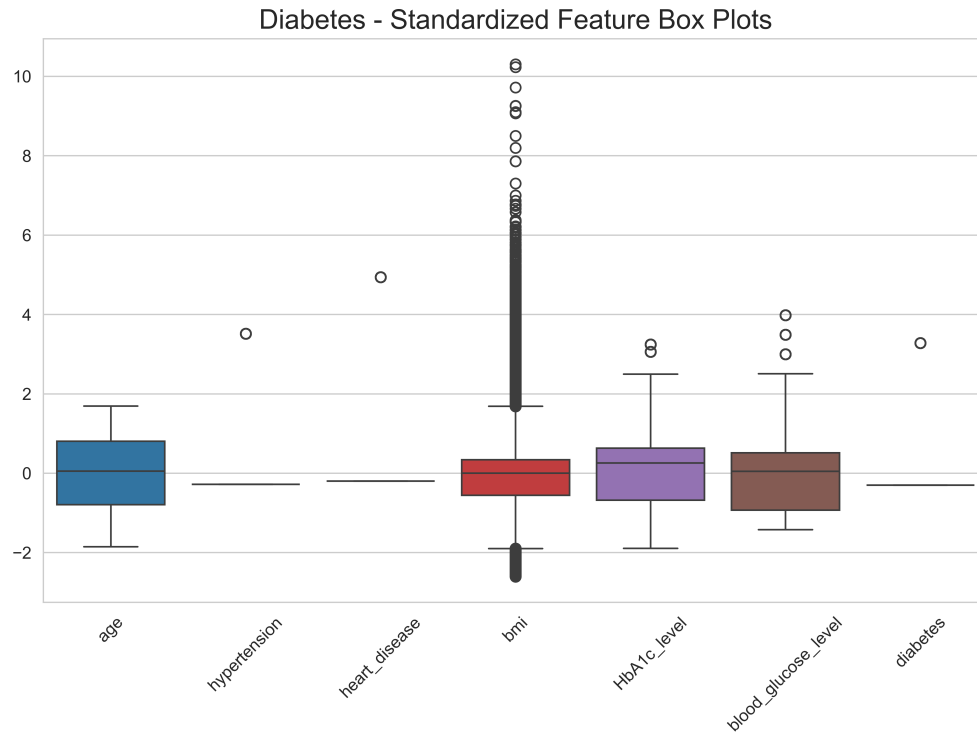


Figure 10: Standardized Box Plots - Diabetes Dataset

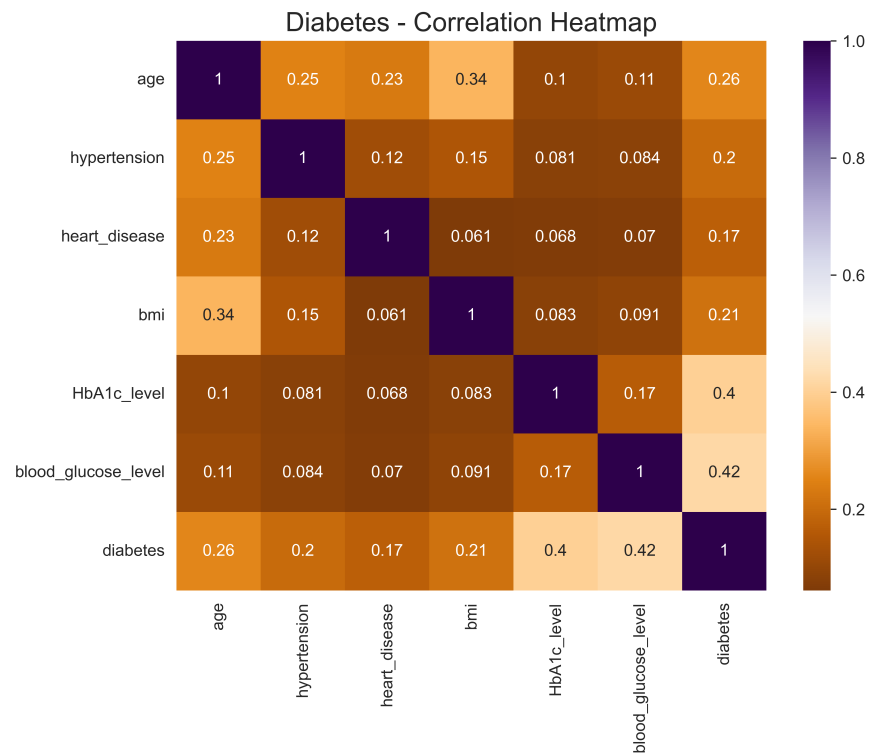


Figure 11: Correlation Heatmap - Diabetes Dataset

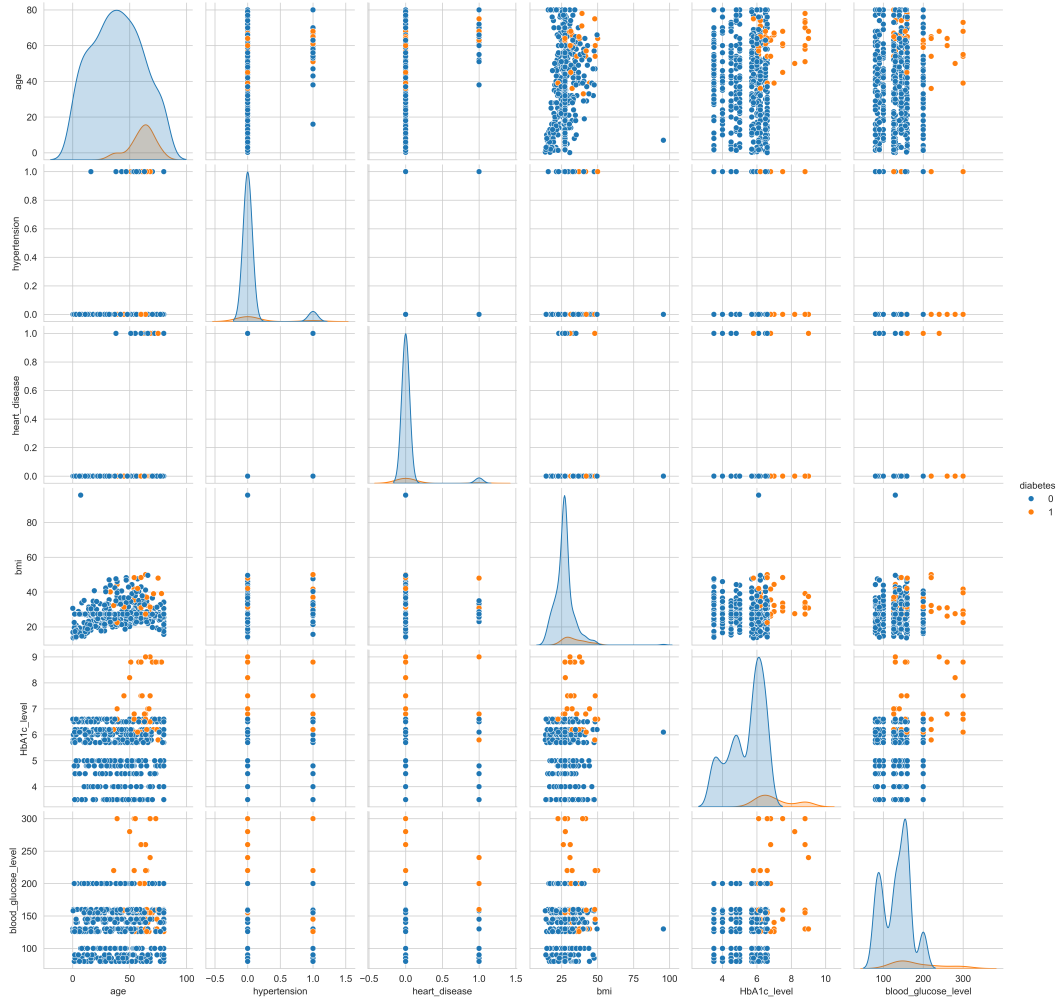


Figure 12: Pair Plot Classification - Diabetes Dataset

3.4 Classification of Email Spam

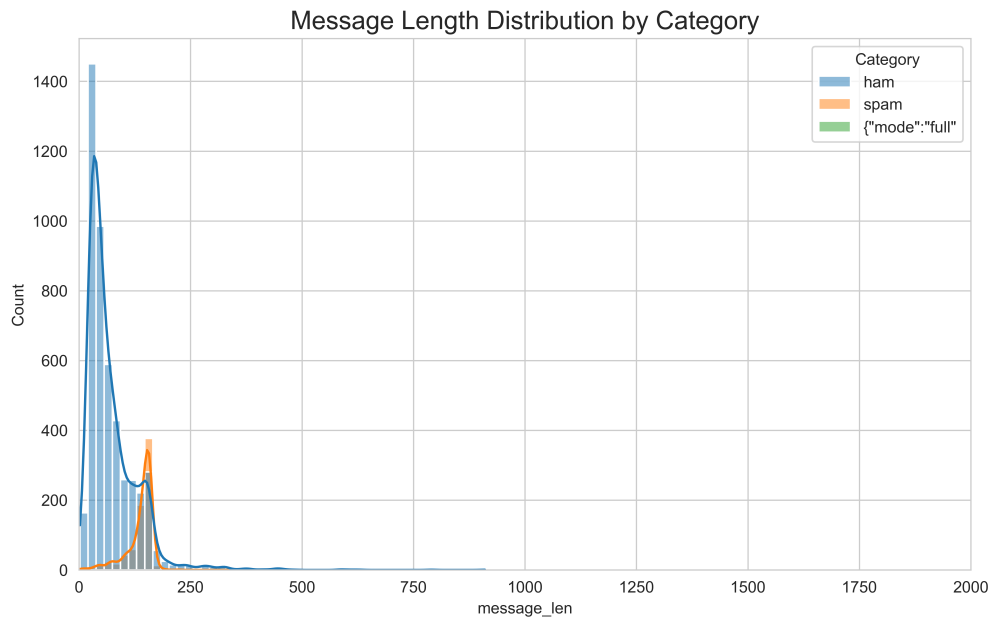


Figure 13: Message Length Distribution - Email Spam

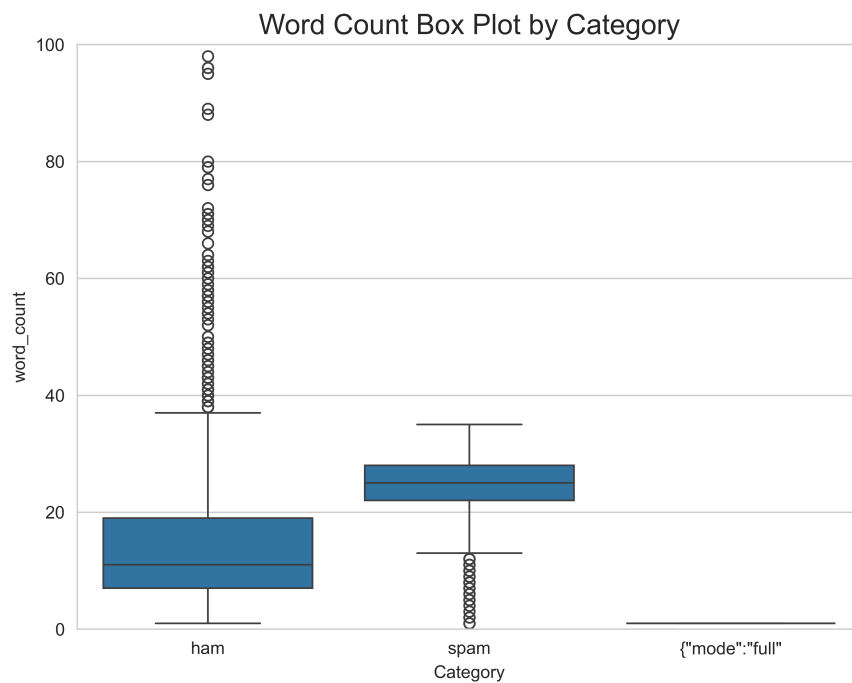


Figure 14: Word Count Box Plot - Email Spam

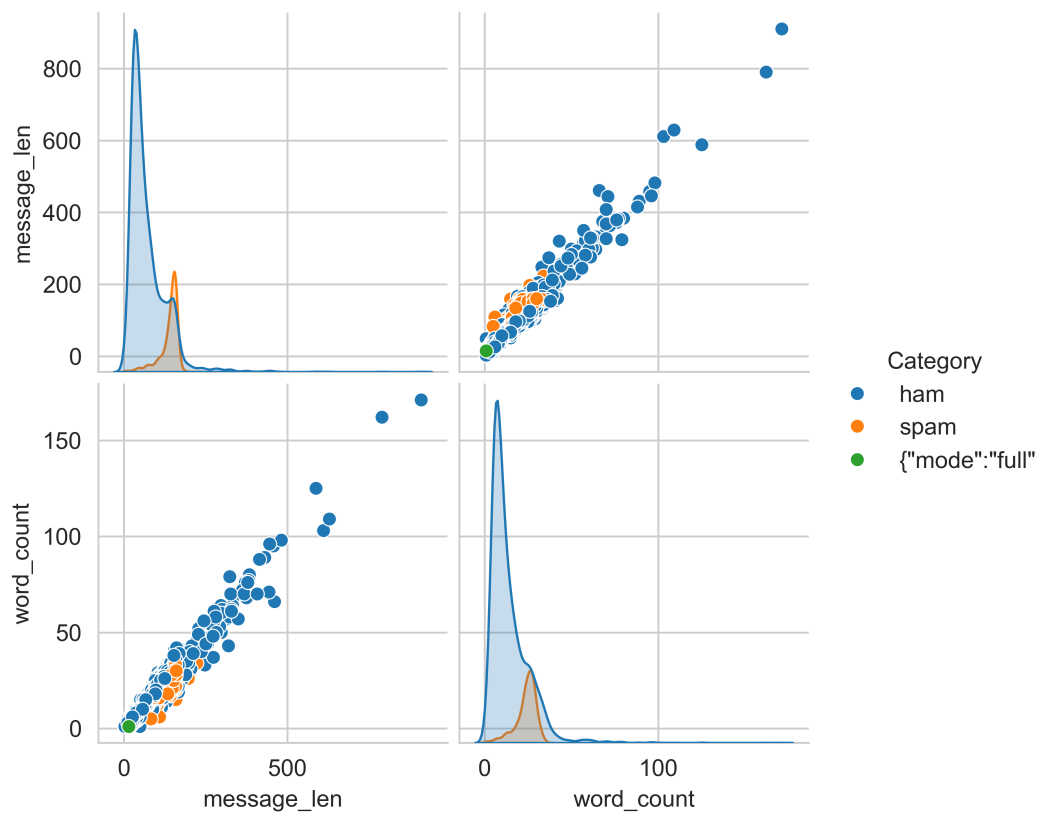


Figure 15: Pair Plot Analysis - Email Spam

3.5 Handwritten Character Recognition / MNIST

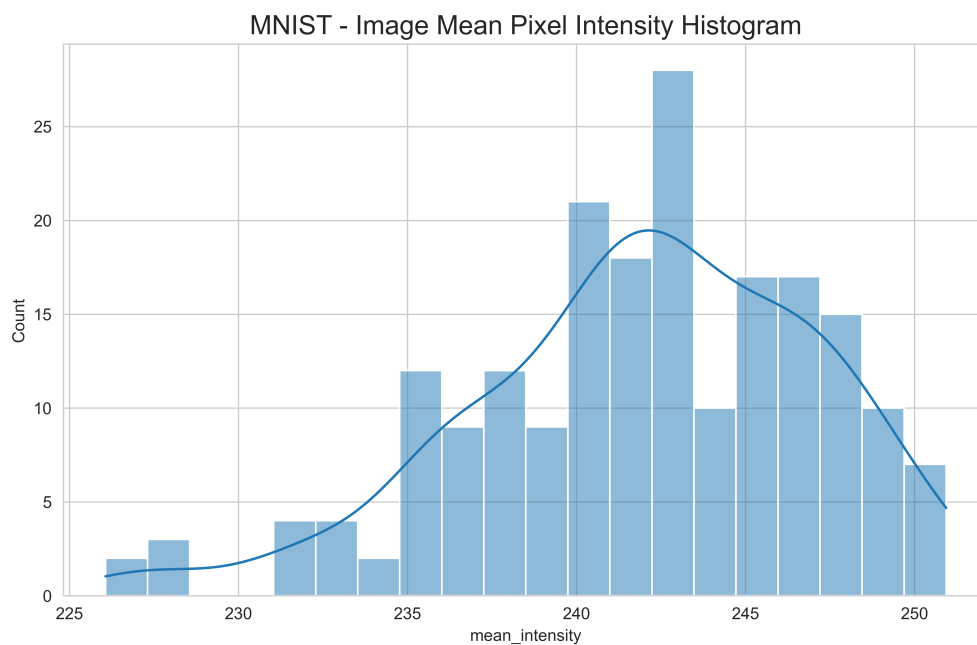


Figure 16: Pixel Intensity Histogram - MNIST

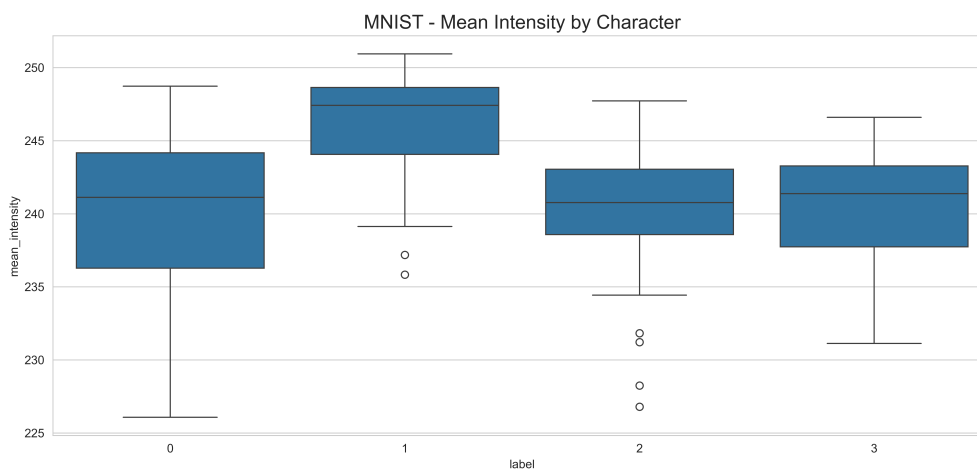


Figure 17: Mean Intensity Box Plot - MNIST

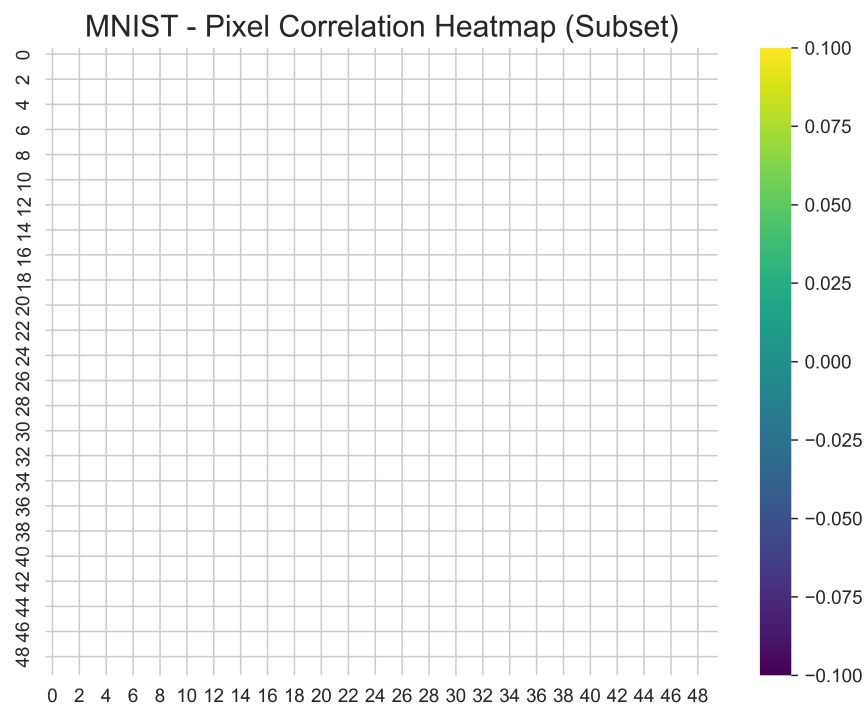


Figure 18: Pixel Correlation Heatmap (Subset) - MNIST

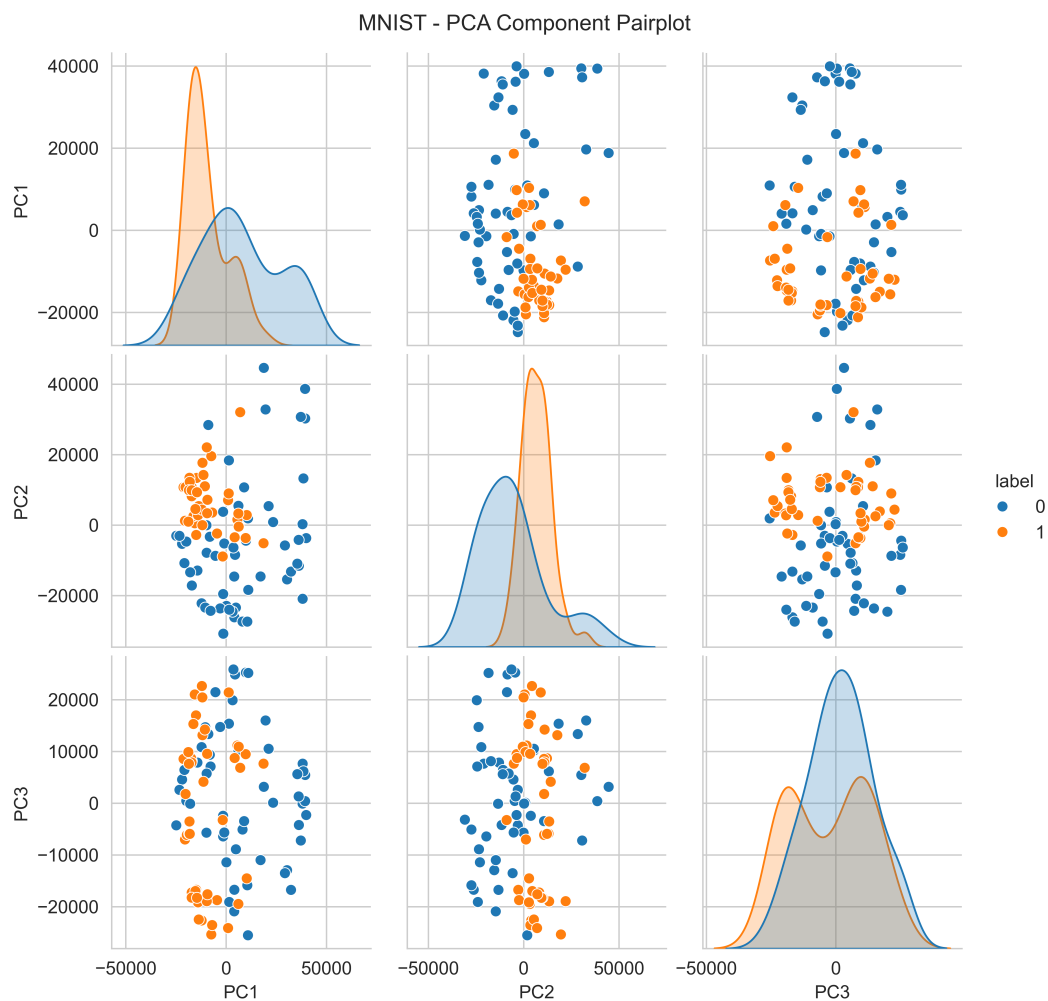


Figure 19: PCA Components Pair Plot - MNIST

4. ML Task Analysis

Table 1: Machine Learning Task Analysis for Each Dataset

Dataset	Type of ML Task	Feature Selection Technique	Suitable ML Algorithm
Iris Dataset	Multi-class Classification	Correlation Analysis, PCA, Variance Analysis	Logistic Regression, Decision Tree, K-Nearest Neighbors, Support Vector Machine
Loan Amount Prediction	Regression	Correlation Matrix, Feature Importance, Variance Inflation Factor	Linear Regression, Ridge Regression, Random Forest Regressor, Gradient Boosting
Predicting Diabetes	Binary Classification	Chi-square Test, Mutual Information, Recursive Feature Elimination	Logistic Regression, Random Forest, Support Vector Machine, Naive Bayes
Classification of Email Spam	Binary Classification	TF-IDF Vectorization, Chi-square Test, Information Gain	Naive Bayes, Support Vector Machine, Logistic Regression, Random Forest
Handwritten Character Recognition / MNIST	Multi-class Classification	Principal Component Analysis, Variance Threshold, Feature Extraction	Convolutional Neural Network, Random Forest, Support Vector Machine, K-Nearest Neighbors

5. Key Observations

5.1 Data Loading and Preprocessing

- Successfully loaded all datasets using appropriate Python libraries including Pandas for structured data and sklearn.datasets for benchmark datasets
- Systematically identified and quantified missing values across all datasets and implemented appropriate imputation strategies
- Performed necessary data type conversions to ensure compatibility with machine learning algorithms
- Applied standardization and normalization techniques to numerical features to ensure uniform scale and improve model convergence
- Handled categorical variables through appropriate encoding techniques where applicable

5.2 Exploratory Data Analysis

- Conducted comprehensive analysis of class distributions to identify potential imbalances that may require sampling techniques
- Examined inter-feature correlations using correlation matrices and identified multicollinearity issues
- Detected outliers using multiple statistical methods including box plots, Z-scores, and interquartile range analysis
- Visualized feature distributions through histograms, kernel density plots, and quantile-quantile plots to assess normality
- Investigated relationships between features and target variables using scatter plots and pair plots
- Analyzed feature importance and contribution to model predictions

5.3 ML Task Identification

- Accurately distinguished between classification and regression tasks based on target variable characteristics and problem statements
- Differentiated between binary and multi-class classification problems and their respective evaluation metrics
- Recommended appropriate machine learning algorithms considering dataset size, dimensionality, and computational constraints
- Identified suitable feature selection and dimensionality reduction techniques aligned with specific task requirements
- Considered algorithm interpretability requirements for different application domains
- Evaluated trade-offs between model complexity and performance for each dataset

6. Conclusion

This comprehensive experiment provided extensive hands-on experience with essential Python libraries that form the foundation of modern machine learning workflows. Through systematic loading, preprocessing, exploration, and visualization of five diverse datasets spanning various domains, valuable insights were gained into different machine learning task types and their corresponding algorithmic approaches. The exploratory data analysis revealed critical patterns, correlations, and statistical characteristics that directly inform algorithm selection, hyperparameter tuning, and preprocessing strategies. The experiment successfully demonstrated the importance of understanding data characteristics before model development and highlighted the interconnected nature of data preprocessing, feature engineering, and algorithm selection in building effective machine learning solutions.

References

- Numpy – Official Documentation: <https://numpy.org/doc/>
- Pandas – Official Documentation: <https://pandas.pydata.org/docs/>
- Scikit-learn – Official Documentation: <https://scikit-learn.org/stable/>

- Matplotlib – Official Documentation: <https://matplotlib.org/stable/contents.html>
- Scipy – Official Documentation: <https://docs.scipy.org/doc/>
- UCI Machine Learning Repository: <https://archive.ics.uci.edu/ml/>
- Kaggle Datasets: <https://www.kaggle.com/datasets>